

REACT

01 FUNDAMENTALS AND INSTALLATION

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SYLLABUS

- **01 Fundamentals And Installation**
- 02 Components And Props
- 03 State
- 04 Handling Events
- 05 SetState And Lifecycle
- 06 Lists And Keys
- 07 Forms
- 08 PropTypes
- 09 Refs
- 10 React Router v4

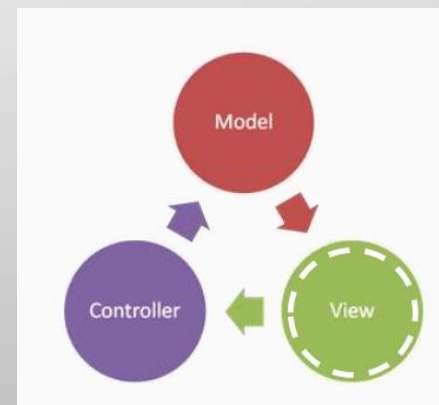
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מה זה REACT?

React

A JavaScript library for building user interfaces

- ספריה לפיתוח UI של צד לקוח בJS.



- V מתוך MVC

- DECLERATIVE - מתכנתים את המה רוצים לעשות? ולא את האיך משיגים את זה?

Declarative

A tower of 3 blocks

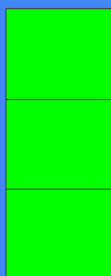


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Imperative

1. Put down block A
2. Put down block B on A
3. Put down block C on B



Learn once, write anywhere? מה זה REACT

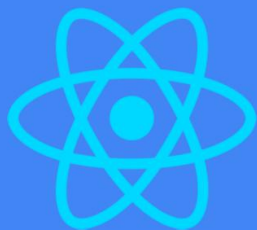
Write once, run anywhere



It's Java



React Native



React VR



LEARN ONCE, •
WRITE ANYWHERE

– משתלב עם
טכנולוגיות אחרות,
רץ גם בשרת, יש גם
REACT ANTIVE

הכל זה COMPONENT •



מי משתמש?



כולל
ANGULAR 2

פופולריות

angular

One framework. Mobile & desktop.

● TypeScript ★ 29.6k 🍴 7.4k

angular.js

AngularJS - HTML enhanced for web apps!

● JavaScript ★ 57.4k 🍴 28.7k

react

A declarative, efficient, and flexible JavaScript interfaces.

frontend declarative ui library

● JavaScript ★ 80,221 🍴 15,201

react-native

A framework for building native apps

● JavaScript ★ 55,430 🍴 12,928

vue

A progressive, incrementally-adoptable JavaScript framework for building UI on the web.

● JavaScript ★ 72.6k 🍴 10.6k



Search GitHub

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Gitstar Ranking

Users

Organizations

Repositories

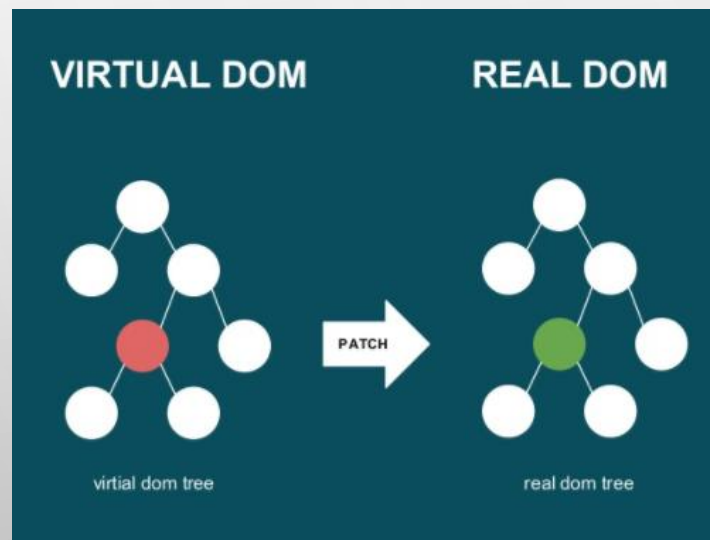
GitHub username

Repositories Ranking

1.	freeCodeCamp/freeCodeCamp	★ 291193
2.	twbs/bootstrap	★ 114834
3.	EbookFoundation/free-progra...	★ 92555
4.	facebook/react	★ 74873
5.	tensorflow/tensorflow	★ 68969
6.	d3/d3	★ 67875
7.	getify/You-Dont-Know-JS	★ 66736
8.	vuejs/vue	★ 65590
9.	sindresorhus/awesome	★ 64991
10.	robbyrussell/oh-my-zsh	★ 58591
11.	airbnb/javascript	★ 57906
12.	angular/angular.js	★ 56966
13.	github/gitignore	★ 54224
14.	facebook/react-native	★ 52994

מרכיבים עיקריים

• VIRTUAL DOM



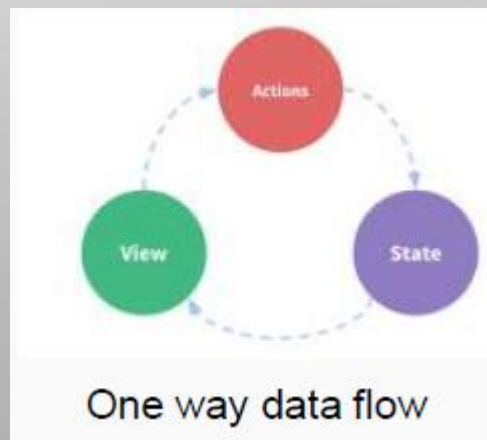
• JSX

• הכל זה COMPONENTS



Components

• המידע זורם בכיוון אחד



One way data flow

מרכיבים עיקריים

- הכי מהירה בצד לקוח (יותר מANGULAR, AMBER...) בגלל הVIRTUAL DOM. FPS 60



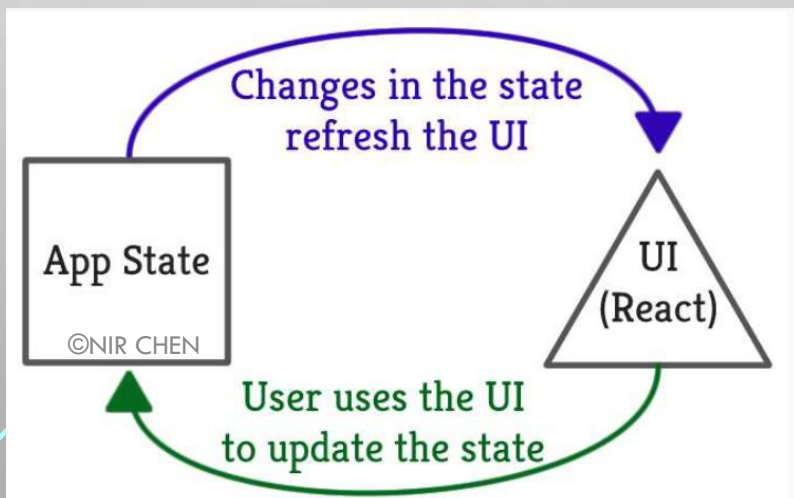
DOM VS Virtual DOM



VS



- הכנסת ה"HTML" (JSX) לתוך הJS לעומת שפות אחרות שמכניסות את ה JS לתוך ה HTML. (נראה בהמשך)
- משנים את המצב של התוכנית והיא מעדכנת אוטומטית את הUI.



עזרים

BABEL

Babel is a JavaScript compiler.

Use next generation JavaScript, today.

- BABEL – מתרגם JSX לJS.

יודע גם לתרגם מES6 לJS ישן

כך שיעבוד בדפדפנים ישנים

npm

npm is the package manager for JavaScript and the world's largest software registry. Discover packages of reusable code — and assemble them in powerful new ways.

Use npm to install, share, and distribute code; manage dependencies in your projects; and share & receive feedback with others.

- NPM – עוזר לארגן את הפרוייקט

להוריד ספריות ולייצר גרסת הרצה

עזרים



At its core, *webpack* is a *module bundler* for modern JavaScript applications. When webpack processes your application, it recursively builds a *dependency graph* that includes every module your application needs, then packages all of those modules into one or more *bundles*.

• WEBPACK – מייצר קבצים להרצה כך שיהיו מאוחדים, וקטנים יותר ללא כפילויות.



Linting tools like ESLint allow developers to discover problems with their JavaScript code without executing it.

• מעיין קומפיילר ל JS ו- JSX



• סביבת העבודה שלנו תהיה

INSTALL

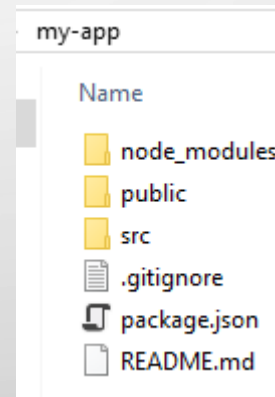
- Nodejs (containing npm)
- Git



הרצה

FOR DEVELOPMENT

```
npx create-react-app my-app  
cd my-app  
npm start / npm run build
```



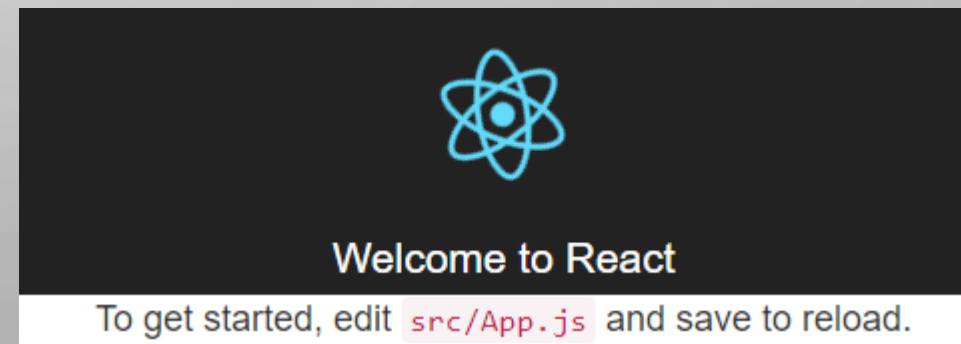
FOR PRODUCTION

```
npx create-react-app my-app  
cd my-app
```

```
npm run build
```

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```
npm install -g serve  
Serve -s build
```



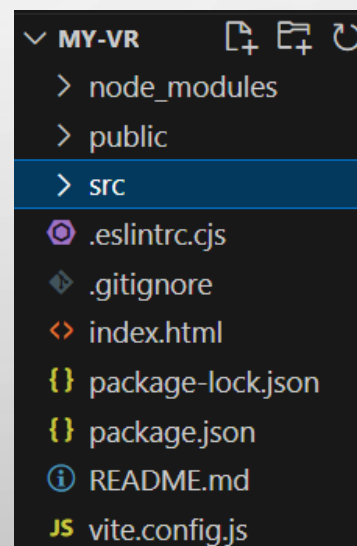
VITE - FRONTEND TOOLING

FOR DEVELOPMENT

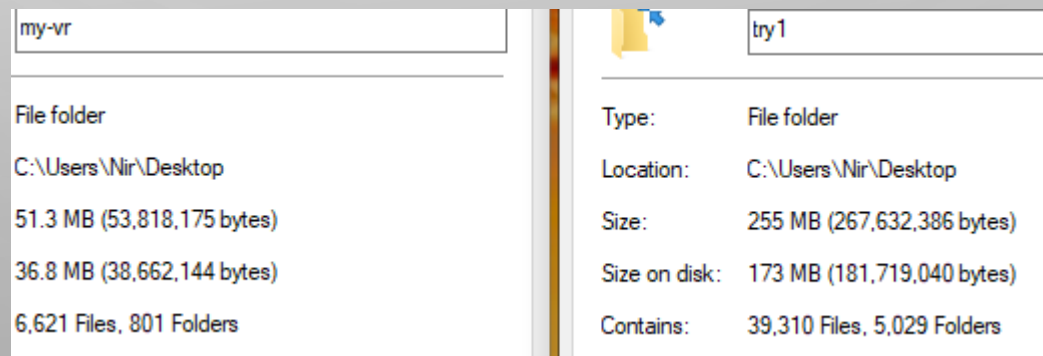
```
npm create vite@latest  
cd my-app  
npm run dev
```

FOR PRODUCTION

```
npm run build
```



vite



Create-react-app

RUN ON SERVER

FOR PRODUCTION

```
npx create-react-app my-app  
cd my-app  
npm start
```

npm run build



להעתיק את
ספריית BUILD
לשרת



package.json

```
{  
  "name": "my-app",  
  "version": "0.1.0",  
  "private": true,  
  "dependencies": {  
    "react": "^16.0.0",  
    "react-dom": "^16.0.0",  
    "react-scripts": "1.0.17"  
  },  
  "scripts": {  
    "start": "react-scripts start",  
    "build": "react-scripts build",  
    "test": "react-scripts test --env=jsdom",  
    "eject": "react-scripts eject"  
  },  
  "homepage":  
    "http://ruppinmobile.ac.il.prev.../site01/build"  
}
```


RUN ON SERVER - VITE

FOR PRODUCTION

```
npx create-react-app my-app
cd my-app
npm start
```

```
npm run build
```



הפקודה תייצר לכם ספריית DIST ואז חובה להכנס עליה ולשנות בקובץ index.html את הלינקים של הJS וה CSS ולהוסיף להם נקודה כך שיתחילו ב `"/assests..."`

לשים את התמונות בספריית public בהנחה שהתמונה יושבת ישירות ב public src="./img.jpg"

```
<meta charset="UTF-8" />
<link rel="icon" type="image/svg+xml" href="/vite.svg" />
<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<title>Vite + React</title>
<script type="module" crossorigin src="/assets/index-01c2c680.js"></script>
<link rel="stylesheet" href="/assets/index-d526a0c5.css">
</head>
<body>
```

```
const router = createHashRouter([
  {
    path: "/",
    element: <Root />,
    //loader: rootLoader,
    children: [
```

שלב ראשון
לייצר את הקובץ
הזה

.eslintrc.js

```
module.exports = {  
  "parser": "babel-eslint",  
  // "extends": "airbnb",  
  "extends": ["eslint:recommended",  
    "plugin:react/recommended"],  
  "parserOptions": {  
    "ecmaFeatures": {  
      "jsx": true,  
      "modules": true  
    }  
  },  
  "plugins": [  
    "react",  
    "jsx-a11y",  
    "import"  
  ],  
};
```

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ESLINT

שלב שני

```
npm install eslint --save-dev
```

```
npm install eslint-plugin-react --save-dev
```

```
npm install -g install-peerdeps  
install-peerdeps --dev eslint-config-airbnb
```



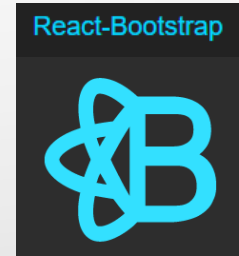
Linting tools like ESLint allow developers to discover problems with their JavaScript code without executing it.

BOOTSTRAP

```
npm install --save bootstrap
```

index.js

```
import '../node_modules/bootstrap/dist/css/bootstrap.min.css';
```



• מתי צריך את react-bootstrap

- If you just need html and css part of bootstrap, it should be fine to not use the library .But if you want use js part of bootstrap for bootstrap transitions, the library is a must

TRY IT YOURSELF



Welcome to React

To get started, edit `src/App.js` and save to reload.

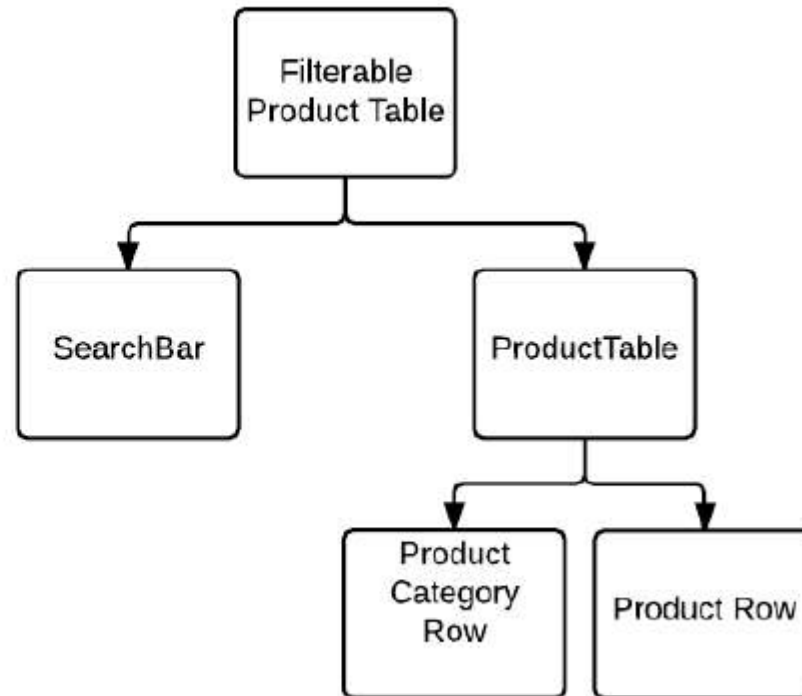
Hello, world!

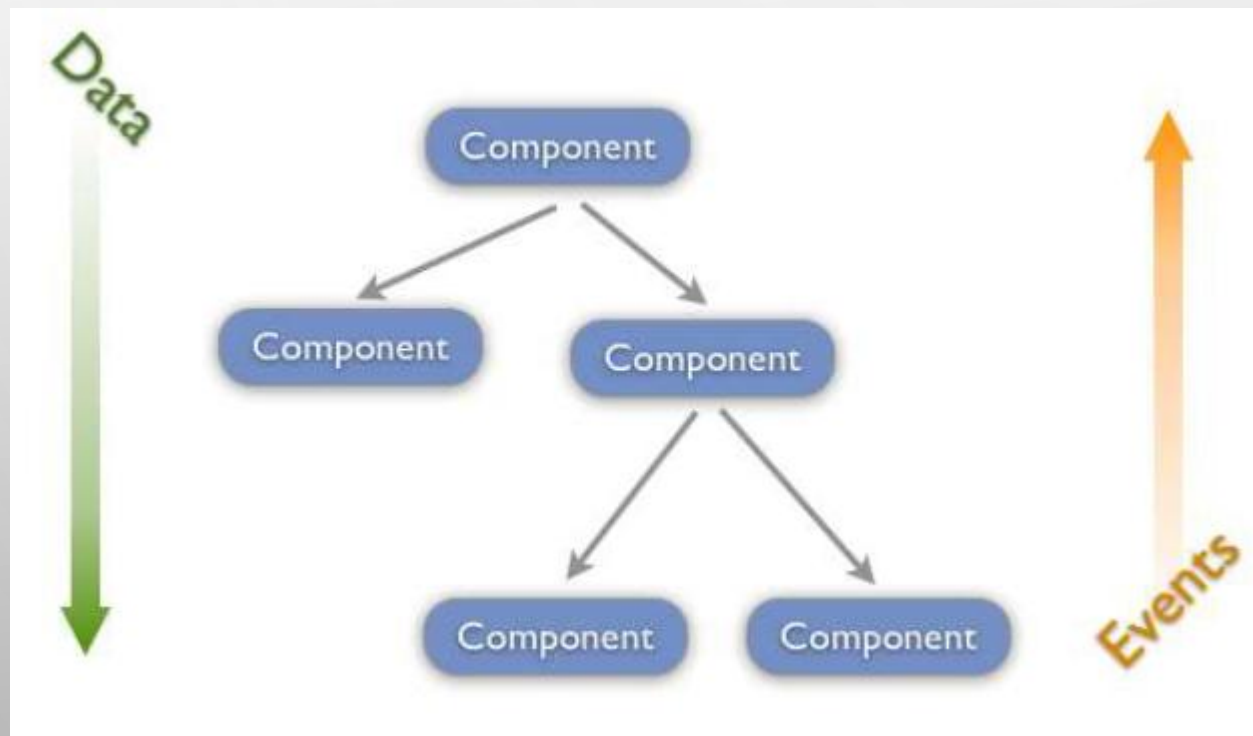
From our React Course:)

COMPONENTS

☒ Only show products in stock

Name	Price
Sporting Goods	
Football	\$49.99
Baseball	\$9.99
Basketball	\$29.99
Electronics	
iPod Touch	\$99.99
iPhone 5	\$399.99
Nexus 7	\$199.99





Mounting

initialization
(initial **state** or default **props**)

componentWillMount

render

componentDidMount

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Updating

componentWillReceiveProps

shouldComponentUpdate

true  false

componentWillUpdate

render

componentDidUpdate

Unmounting

componentWillUnmount